CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech (Electronics & Communication) SEM - IV University Examination, 2010-11

EC 209 Microprocessor and Microcontroller

Date: 10.05.11, Tuesday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION –I

Q-1 Answer the following

(a) Draw the block diagram of 8085 microprocessor and discuss Program status word in detail — 4

(b) What is memory Mapped I/O? discuss this technique in comparison with peripheral mapped I/O. Give ILLUSATRATION of both technique by interfacing example — 4

(c) Give the comparison of following Microprocessor and Microcontroller: 8051,8031 and 8085 — 3

Q-2(a) Draw the interfacing diagram of 8085 microprocessor with 64Kx8 EPROM memory. Consider a system in which the full memory space is utilized. Mentioned the address range of EPROM. — 6

(b) Explain the block diagram of Programmable Timer / counter device and List out the different types of modes used with device. — 6

OR

Q-2(a) Make a tabular comparison of Partial decoding and Absolute decoding. Discuss any ONE with an interfacing example. — 6

(b) Draw the block diagram of Programmable Peripheral Interface device and explain the BSR word format with example — 6

Q-3(a) Draw the Ceramic Resonator circuit with 8051 and explain One Machine cycle concept relevance with ALE pulse — 6

(b) Draw the SFR format of TMOD and TCON. Explain both with necessary example to SET /RESET the bit and Write instruction to initialize SFR — 6

OR

Q-3(a) Discuss the importance of pull-up resistors and Draw the Interfacing diagram to connect 8 LED with AT89C51 on port P0. — 6

(b) Draw the SFR format of SBUF and SCON. Explain both with necessary example to SET /RESET the bit and Write instruction to initialize SFR — 6

SECTION-II

Q-4	(a)	What is addressing modes for 8051 microcontroller? List out all and explain with necessary examples	6
	(b)	Discuss the importance of Programming tools. Make a table to compare following tools : - Assembler - Compiler - Cross-compiler	5
Q-5	(a)	The ON/OFF switches are connected at Port P3. Read the data form Port P1 and send the squared value of data to Port P2 .Write a code for 8051.	6
	(b)	List out the types of Boolean logic instructions and explain with examples. Also discuss the effect on PSW by each type.	6
		OR	
Q-5	(a)	Write an assembly language program for 8051 to make a subroutine that will generate delay of exact 1 ms. Use this delay to generate square wave of 50 Hz on pin P2.0. The crystal frequency is given as 12 MHz. Show necessary calculations for generating delay	6
	(b)	What is Interrupt? Write a program to count number of interrupts arriving on external interrupt pin INT0 of 8051. Stop when counter overflows and disable the interrupt. Give the indication on pin P1.2 where LED is connected.	6
Q-6		**Write short note on (Any Three)**	12
	(a)	Opcode Fetch cycle	
	(b)	Jump and Call Program Range	
	(c)	8051 Data Communication Modes	
	(d)	Lookup Table techniques for 8051	
	(e)	8051 Internal Architecture	